GRAPH LAPLACEANS

- Graph Laplaceans, definitions
- Properties of graph properties
- Graph partitioning –

Graph Laplaceans - Definition

- "Laplacean-type" matrices defined for general undirected graphs

Definition

- Given a graph $G = (V, E)$ define
  - A matrix W of weights w_{ij} for each edge
  - Assume $w_{ij} \geq 0$,, $w_{ii} = 0$, and $w_{ij} = w_{ji}\ \forall (i,j)$
  - The diagonal matrix $D = diag(d_i)$ with $d_i = \sum_{j \neq i} w_{ij}$
- Corresponding *graph Laplacean* of G is: $\boxed{L = D - W}$

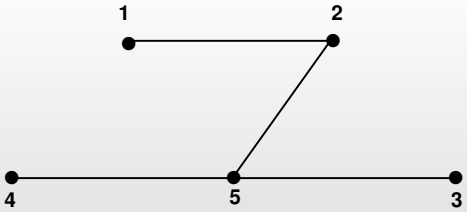
- Gershgorin's theorem $\rightarrow L$ is positive semidefinite.

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- Simplest case:

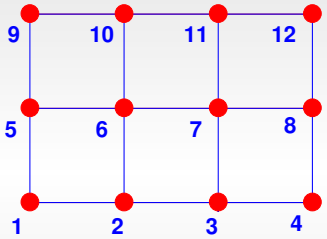
$$w_{ij} = \begin{cases} 1 & \text{if} \quad (i,j) \in E \;\&\; i \neq j \\ 0 & \text{else} \end{cases} \qquad D = \text{diag}\left[d_i = \sum_{j \neq i} w_{ij} \right]$$

Example: Consider the graph



$$L = \begin{bmatrix} 1 & -1 & 0 & 0 & 0 \\ -1 & 2 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 1 & -1 \\ 0 & -1 & -1 & -1 & 3 \end{bmatrix}$$

✍ 1 Define the graph Laplacean for the graph associated with the simple mesh shown next. [use the simple weights of 0 or 1]. What is the difference with the discretization of the Laplace operator for case when mesh is the same as this graph?



Proposition:

(i) $\boldsymbol{L}$ is symmetric semi-positive definite.

(ii) $\boldsymbol{L}$ is singular with $\mathbb{1}$ as a null vector.

(iii) If $\boldsymbol{G}$ is connected, then $\textbf{Null}(\boldsymbol{L}) = \textbf{span}\{\mathbb{1}\}$

(iv) If $\boldsymbol{G}$ has $\boldsymbol{k} > \mathbf{1}$ connected components $\boldsymbol{G_1}, \boldsymbol{G_2}, \cdots, \boldsymbol{G_k}$, then the nullity of $\boldsymbol{L}$ is $\boldsymbol{k}$ and $\textbf{Null}(\boldsymbol{L})$ is spanned by the vectors $z^{(j)}$, $\boldsymbol{j} = \mathbf{1}, \cdots, \boldsymbol{k}$ defined by:

$$(z^{(j)})_i = \begin{cases} \mathbf{1} & \text{if } \boldsymbol{i} \in \boldsymbol{G_j} \\ \mathbf{0} & \text{if not.} \end{cases}$$

- (Note: $\mathbb{1}$ == vector of all ones)

Proof: (i) and (ii) seen earlier and are trivial. (iii) Clearly $\boldsymbol{u} = \mathbb{1}$ is a null vector for $\boldsymbol{L}$. The vector $\boldsymbol{D}^{-1/2}\boldsymbol{u}$ is an eigenvector for the matrix $\boldsymbol{D}^{-1/2}\boldsymbol{L}\boldsymbol{D}^{-1/2} = \boldsymbol{I} - \boldsymbol{D}^{-1/2}\boldsymbol{W}\boldsymbol{D}^{-1/2}$ associated with the smallest eigenvalue. It is also an eigenvector for $\boldsymbol{D}^{-1/2}\boldsymbol{W}\boldsymbol{D}^{-1/2}$ associated with the largest eigenvalue. By the Perron Frobenius theorem this is a simple eigenvalue... (iv) Can be proved from the fact that $\boldsymbol{L}$ can be written as a direct sum of the Laplacian matrices for $\boldsymbol{G_1}, \cdots, \boldsymbol{G_k}$. ■

Define: oriented incidence matrix H

(1)First orient the edges $i \sim j$ into $i \to j$ or $j \to i$. (2) Rows of H indexed by vertices of G. Columns indexed by edges. (3) For each (i,j) in E, define the corresponding column in H as $\sqrt{w(i,j)}(e_i - e_j)$.

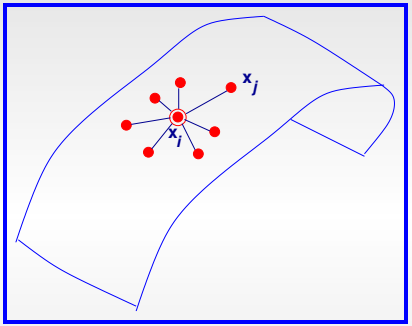
Example: In previous example (P. 4) orient $i \to j$ so that $j > i$ [lower triangular matrix representation]. Then matrix H is:

$$H = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & -1 & -1 & -1 \end{bmatrix}$$

Property 1 $\boxed{L = HH^T}$

✍ 2 Re-prove part (iv) of previous proposition by using this property.

A few properties of graph Laplaceans



- Strong relation between $x^T Lx$ and local distances between entries of x
- Let L = any matrix s.t. $L = D - W$, with $D = diag(d_i)$ and

$$w_{ij} \geq 0, \qquad d_i = \sum_{j \neq i} w_{ij}$$

Property 2: for any $x \in \mathbb{R}^n$:

$$x^\top Lx = \frac{1}{2} \sum_{i,j} w_{ij} |x_i - x_j|^2$$

Property 3: (generalization) for any $Y \in \mathbb{R}^{d \times n}$:

$$\text{Tr}\,[YLY^\top] = \frac{1}{2}\sum_{i,j} w_{ij}\|y_i - y_j\|^2$$

✦ Note: $y_j = j$-th colunm of Y. Usually $d < n$. Each column can represent a data sample.

Property 4: For the particular $L = I - \frac{1}{n}\,\mathbb{1}\,\mathbb{1}^\top$

$$XLX^\top = \bar{X}\bar{X}^\top == n \times \text{Covariance matrix}$$

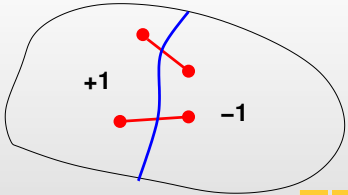
Property 5: L is singular and admits the null vector $\mathbb{1} \equiv$`ones(n,1)`

Property 6: (Graph partitioning) Consider situation when $w_{ij} \in \{0, 1\}$. If x is a vector of signs (± 1) then

$$x^\top L x = 4 \times \text{('number of edge cuts')}$$

edge-cut = pair (i, j) with $x_i \neq x_j$

- Consequence: Can be used to partition graphs



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- Would like to minimize (Lx, x) subject to $x \in \{-1, 1\}^n$ and $e^T x = 0$ [balanced sets]
- WIl solve a relaxed form of this problem

✍ 3 What if we replace x by a vector of ones (representing one partition) and zeros (representing the other)?

✍ 4 Let x be any vector and $y = x + \alpha \mathbb{1}$ and L a graph Laplacean. Compare (Lx, x) with (Ly, y).

- Consider any symmetric (real) matrix A with eigenvalues $\lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_n$ and eigenvectors $u_1, \cdots, u_n$
- Recall that:
(Min reached for $x = u_1$)

$$\min_{x \in \mathbb{R}^n} \frac{(Ax, x)}{(x, x)} = \lambda_1$$

- In addition:
(Min reached for $x = u_2$)

$$\min_{x \perp u_1} \frac{(Ax, x)}{(x, x)} = \lambda_2$$

- For a graph Laplacean $u_1 = \mathbb{1}$ = vector of all ones and
- ...vector u_2 is called the Fiedler vector. It solves a relaxed form of the problem -

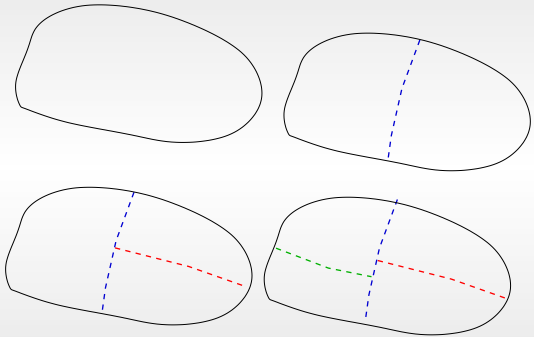
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$$\min_{x \in \{-1,1\}^n;\ \mathbb{1}^T x = 0} \frac{(Lx, x)}{(x, x)} \quad \rightarrow \quad \min_{x \in \mathbb{R}^n;\ \mathbb{1}^T x = 0} \frac{(Lx, x)}{(x, x)}$$

- Define $v = u_2$ then define label$(v) = sign(v - med(v))$

Recursive Spectral Bisection

1. Form graph Laplacean
2. Partition graph in 2 based on Fielder vector
3. Partition largest subgraph in two recursively ...
4. ... Until the desired number of partitions is reached



Three approaches to graph partitioning:

1. Spectral methods - Just seen + add Recursive Spectral Bisection.
2. Geometric techniques. Coordinates are required. [Houstis & Rice et al., Miller, Vavasis, Teng et al.]
3. Graph Theory techniques – multilevel,... [use graph, but no coordinates]
   - Currently best known technique is Metis (multi-level algorithm)
   - Simplest idea: Recursive Graph Bisection; Nested dissection (George & Liu, 1980; Liu 1992]
   - Advantages: simplicity – no coordinates required

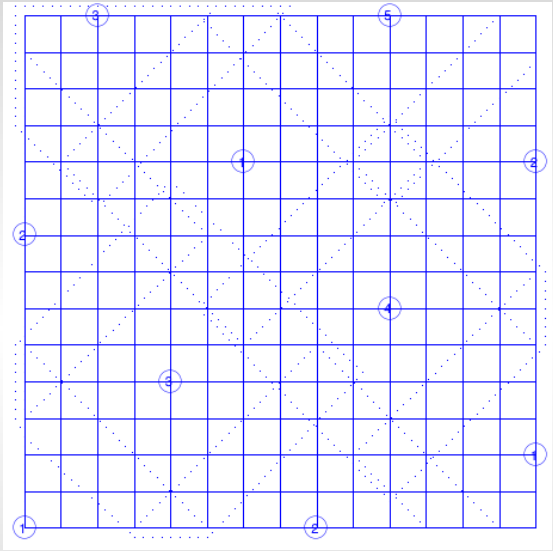
✍ 5 Run *testBis_simple* and *testMeshPart* in class repository

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Example of a graph theory approach

- Level Set Expansion Algorithm
- Given: p nodes 'uniformly' spread in the graph (roughly same distance from one another).
- Method: Perform a level-set traversal (BFS) from each node simultaneously.
- Best described for an example on a 15×15 five – point Finite Difference grid.
- See [Goehring-YS '94, See Cai-YS '95]
- Approach also known under the name 'bubble' algorithm and implemented in some packages [Party, DibaP]

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